Figure 1 — System Architecture: Eight-Stage Integrated Pipeline

COMPONENT LEGEND

- Anomaly Detection / Verification (Isolation Forest · LR re-evaluation)
- Temporal Processing (10-step buffer · 45 features)
- AI Prediction + XAI (Logistic Regression · Coefficient Attribution)
- Decision Intelligence (Evidence Scoring · Rule Engine)
- Simulated Recovery (NOT real cloud orchestration)

All stages implemented in Python/Flask. Recovery actions are simulated string selections, not cloud API calls.
Post-healing verification re-evaluates synthetic normal telemetry using the same trained Logistic Regression model.